

Chain Rule Explanation

The Chain Rule: General Formula

If you have a composite function $f(g(x))$, the derivative is:

$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$$

In words: Derivative of the outer function (evaluated at the inner function) times derivative of the inner function.

Simple Example

Find $\frac{d}{dx}(x^2 + 1)^3$

- Outer function: $f(u) = u^3$, so $f'(u) = 3u^2$
- Inner function: $g(x) = x^2 + 1$, so $g'(x) = 2x$

Apply chain rule:

$$\frac{d}{dx}(x^2 + 1)^3 = 3(x^2 + 1)^2 \cdot 2x = 6x(x^2 + 1)^2$$

Problem 4 Application

In problem 4, we needed to find:

$$\frac{d}{dp} \left[p \cdot u \left(w_0 - \frac{R}{p} \right) \right]$$

This requires both the product rule AND the chain rule.

Step 1: Apply Product Rule

For $\frac{d}{dp}[p \cdot u(w_0 - R/p)]$, use product rule first:

$$\frac{d}{dp}[p \cdot u(w_0 - R/p)] = u \left(w_0 - \frac{R}{p} \right) + p \cdot \frac{d}{dp} \left[u \left(w_0 - \frac{R}{p} \right) \right]$$

Step 2: Apply Chain Rule

Now we need $\frac{d}{dp} \left[u \left(w_0 - \frac{R}{p} \right) \right]$:

- Outer function: $f(w) = u(w)$, so $f'(w) = u'(w)$
- Inner function: $g(p) = w_0 - \frac{R}{p}$, so $g'(p) = \frac{d}{dp} \left(w_0 - \frac{R}{p} \right) = 0 - R \cdot \frac{-1}{p^2} = \frac{R}{p^2}$

By chain rule:

$$\frac{d}{dp} \left[u \left(w_0 - \frac{R}{p} \right) \right] = u' \left(w_0 - \frac{R}{p} \right) \cdot \frac{R}{p^2}$$

Step 3: Combine

Putting it all together:

$$\begin{aligned} \frac{d}{dp} \left[p \cdot u \left(w_0 - \frac{R}{p} \right) \right] &= u \left(w_0 - \frac{R}{p} \right) + p \cdot u' \left(w_0 - \frac{R}{p} \right) \cdot \frac{R}{p^2} \\ &= u \left(w_0 - \frac{R}{p} \right) + u' \left(w_0 - \frac{R}{p} \right) \cdot \frac{R}{p} \end{aligned}$$

Key Insight: The Inner Derivative

The tricky part is finding $\frac{d}{dp} \left(w_0 - \frac{R}{p} \right)$:

$$\begin{aligned} \frac{d}{dp} \left(w_0 - \frac{R}{p} \right) &= \frac{d}{dp} (w_0 - R \cdot p^{-1}) \\ &= 0 - R \cdot (-1) \cdot p^{-2} \\ &= R \cdot p^{-2} \\ &= \frac{R}{p^2} \end{aligned}$$

Why positive? Because $\frac{R}{p}$ decreases as p increases, so $w_0 - \frac{R}{p}$ increases as p increases, giving a positive derivative.

Quick Reference

Common chain rule applications:

$$\begin{aligned} \frac{d}{dx} f(ax + b) &= a \cdot f'(ax + b) \\ \frac{d}{dx} u \left(\frac{1}{x} \right) &= u' \left(\frac{1}{x} \right) \cdot \left(-\frac{1}{x^2} \right) \\ \frac{d}{dx} u \left(\frac{a}{x} \right) &= u' \left(\frac{a}{x} \right) \cdot \left(-\frac{a}{x^2} \right) = u' \left(\frac{a}{x} \right) \cdot \frac{a}{x^2} \end{aligned}$$